3.2 Inorganic chemistry

3.2.1 Periodicity

The Periodic Table provides chemists with a structured organisation of the known chemical elements from which they can make sense of their physical and chemical properties. The historical development of the Periodic Table and models of atomic structure provide good examples of how scientific ideas and explanations develop over time.

3.2.1.1 Classification

Content	Opportunities for skills development
An element is classified as s, p, d or f block according to its position in the Periodic Table, which is determined by its proton number.	

3.2.1.2 Physical properties of Period 3 elements

Content	Opportunities for skills development
The trends in atomic radius, first ionisation energy and melting point of the elements Na–Ar The reasons for these trends in terms of the structure of and bonding in the elements. **Students should be able to:** • explain the trends in atomic radius and first ionisation energy • explain the melting point of the elements in terms of their structure and bonding.	